Alex Li

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Education

Carnegie Mellon University, B.S. Computer Science, Systems Concentration — Aug 2023 – Dec 2026

- **Systems:** Operating Systems, Parallel Computer Architecture and Programming, Distributed Systems, Cloud Computing, Intro to Computer Systems, Intro to Database Systems,
- **AI/ML:** Generative AI, ML with Large Datasets, Intro to Machine Learning, Intro to AI, Intro to Computer Vision
- **Theory & Algorithms:** Parallel and Sequential Data Structures and Algorithms, Functional Programming, Great Ideas in Theoretical CS, Probability, Statistics, Linear Algebra
- **Other CS:** Computer Graphics

Experience

Software Engineering Intern, Sonos — Jun 2026 – Aug 2026

- Rapidly prototyped high-performance, multi-threaded C++ player code for WiFi speakers, enabling new experiences using Bluetooth Low Energy (BLE), WiFi, and ultrasonic signaling
- Engineered distributed communication and state synchronization across Sonos players over WiFi, maintaining coherent views across networked devices
- Developed ultrasonic signal correlation and detection logic to enable reliable interaction between nearby players
- Built an Android app-based emulator for an embedded controller to quickly iterate and test BLE communication and end-to-end device interactions

Software R&D Intern, PTC (Onshape) — Jun 2025 – Aug 2025

- Spearheaded development of an all-new Notes Panel with Markdown support, granular permissions, and preview modes; drove project from design through deployment to all customers in August 2025
- Revamped admin page UI with server-side sorting, pagination, and search in MongoDB for improved scalability
- Practiced agile development and CI/CD best practices within Onshape's three-week sprint cycles

Research Assistant, CMU Search-Based Planning Lab — May 2025 – Aug 2025

- Implemented and evaluated dozens of sampling-based algorithms for single-agent start-to-goal search
- Investigated novel conflict-resolution strategies for multi-agent planning under uncertainty

Robotics & Computer Vision Intern, Dassault Systèmes (Solidworks) — Dec 2024 – Jan 2025

- Designed and developed single- and multi-camera pose estimation (position/orientation) using pure computer vision techniques in Python/C++ with OpenCV. Achieved **5× speedup** and improved precision compared to prior versions

C++ R&D Development Intern, Poisson Software Co. — Jun 2024 – Aug 2024

- Built a high-performance, multi-threaded message queue and broker in C++
- Developed persistent caching to accelerate repeated 3D model computations

Projects

Distributed LZ-Compression on Supercomputer — May 2026

- Wrote custom parallel implementation of LZW and LZ77 compression algorithms designed for Pittsburgh Supercomputer (PSC)
- Used Message Passing Interface (MPI) and OpenMP for CPU-parallelism, and CUDA for GPU-paralellism

In-Car Pedestrian Blind Spot Warning System — May 2026

- Developed a real-time pedestrian detection and blind-spot warning system using camera-based computer vision and embedded hardware mounted in car
- Designed sensor-processing and warning logic to detect pedestrians in vehicle blind spots and provide timely driver alerts through spatial audio and visual interface

B+ Tree and Buffer Pool Manager — Mar 2026

- Implemented a custom Buffer Pool Manager and B+ Tree index for the BusTub DBMS in C++, achieving **top 5** in the Spring 2026 class benchmark.

- Concepts Used: C++, B+ Trees, concurrency control, scalable parallelism, SIMD, caching strategies

Cloud Computing Projects Mar 2025 – Apr 2025

- Created application to elastically scale AWS EC2 instances based on peaks in demand
- Deployed and scaled microservices on Azure using Kubernetes
- Technologies: Java, Python, YAML, AWS, Azure, Kubernetes

TartanAUV - CMU Robotic Submarine Team Aug 2024 – Current

- Learned how the robo-sub uses ROS and refined reinforcement learning models for better object detection/recognition.
- Concepts Used: Python, ROS, reinforcement learning

Malloc Lab Mar 2024

- Wrote custom implementation of Malloc, **Top 3%** in class for S24 in throughput and utilization metrics (74.2%)
- Concepts Used: C, Linked-lists, Memory management

Carnegie Mellon Racing (CMR) Sep 2023 - Current

- Created C++ Delaunay Triangulation midpoint tracing algorithm with lidar data for driverless competition vehicle 22a

Skills

Languages: C++, C, Go (Golang), Rust, Kotlin, Java, Python, JavaScript, TypeScript, Standard ML, HTML/CSS, SQL
Systems/Parallel: Multithreading, Concurrency, Parallel Computing, Distributed Systems, Microservices, OpenMPI, CUDA, TCP/UDP
Frameworks/Tools: Git/GitHub, Android SDK, Docker, Kubernetes, Helm, Angular, REST APIs, PyTorch, NumPy, OpenCV, ROS, Kafka, Spark, Hadoop, Make, CI/CD, GoogleTest
Databases/Cloud: MySQL, MongoDB, Redis, AWS, Azure, Google Cloud, Linux/Unix, Bash

Interests: Ice hockey, piano, saxophone, cars, racing, hiking, climbing, welding